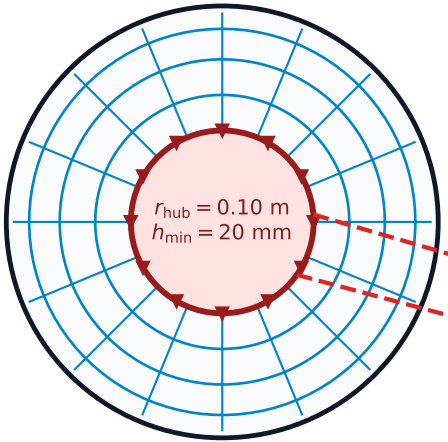
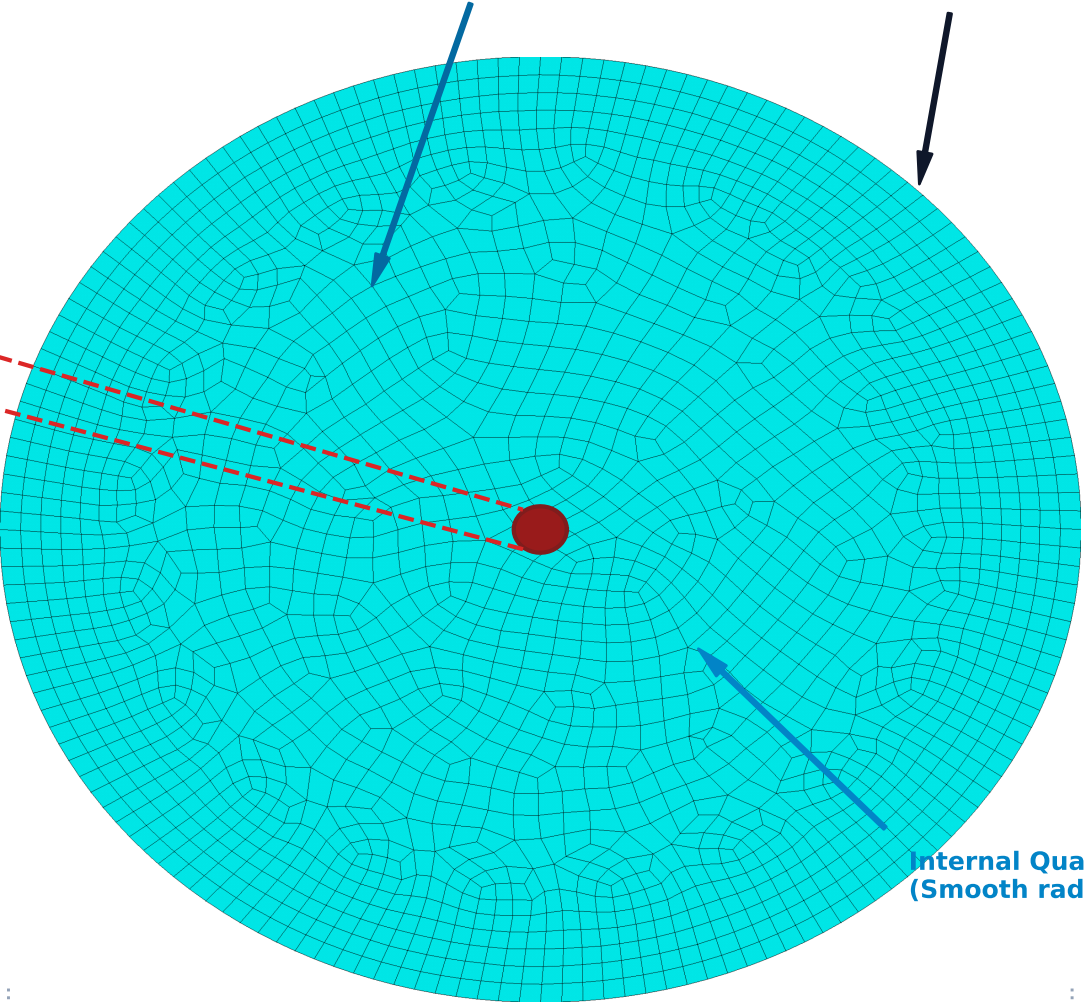


Detail A: Clamped Payload Hub
($UX = UY = UZ = 0$)



Active Reflective Area ($A_{\text{sail}} = 12.57 \text{ m}^2$)

Structured Outer Perimeter Band
($h_{\text{nom}} = 80 \text{ mm}$, **Free Boundary**)



ANSYS FEA Discretization Metrics:

- **Element Type** : SHELL181 (4-Node Structural Shell)
- **Kinematics** : Mindlin-Reissner (Large Defl. NLGEOM)
- **Mesh Density** : 3,048 Elements || 3,112 Nodes
- **Nominal Sizing** : $h_{\text{nom}} = 80 \text{ mm}$ ($h_{\text{min}} = 20 \text{ mm}$)
- **Mesh Quality** : Min Orthog. Qual. = 0.942 (Avg AR = 1.12)

Internal Quad-Dominant Core
(Smooth radial-azimuthal pave transition)

Sail Diameter : D = 4.00 m (Span 2R = 4.00 m)